

CH365 Chemical Engineering Thermodynamics

Lesson 20 Review

Lesson 21 – Thursday 15 October, 55 minutes.

Coverage – Lessons 10-20, Chapters 3 and 4, and Problem Sets 4, 5, 6, and 7.

Problems – A-50, B-70, C-80 pts, 200 pts total.

Calculations – All WPR2 Mathematica uploaded to Canvas.

Ideal gas ΔH , ΔU , W , and Q .

- a. Multi-step processes.
- b. Sketching process paths on PV axes.
- c. Real gases and cubic equations of state.

Real gas and cubic equations of state.

- a. Solving RK, SRK, PR, vdW (virial and Lee-Kesler not on WPR).
- b. Application of cubic equations to calculation of work.

Heat effects.

- a. Sensible heat ($\int C_p dT$, ICPH, MCPH).
- b. Latent heat ($\Delta H_{f,298}^\circ$, ΔH_{298}°).
- c. Heat duty in reactors nonstandard T (IDCPH, MDCPH, $Q = \Delta H_T^\circ$).

Suggestions (Practice exam in Canvas).

- a. Working MMA files for 3.44 (c, d, e), 3.58 (b, c), 4.45, 4.55, and ex. 4-7.
- b. Replicate approved solutions exactly or you have something wrong.